FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

CLOUD COMPUTING (BSE-V)

BS(SE) – Semester V

ASSIGNMENT 01

Git, Gitea, GitHub, Git LFS, and GitHub Pages

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1. Introduction and Objective

This report documents the practical work completed for Assignment 01 of Cloud Computing (BSE-V). In this assignment, I installed and ran a self-hosted Gitea service on an Ubuntu server using Docker Compose, created and managed a local Git repository on the same server, pushed it to both Gitea and GitHub using multiple remotes, tracked large files using Git LFS, and published a portfolio website using GitHub Pages.

By completing this assignment, I demonstrated the ability to:

- Install, run, and access a Gitea service on an Ubuntu server.
- Create and manage a Git repository from an Ubuntu server.
- Work with multiple Git remotes.
- Authenticate safely without exposing credentials.
- Track and push large files with Git LFS.
- Publish a static portfolio or CV with GitHub Pages.

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Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Task meaning: Install Gitea (a self-hosted Git service) on the Ubuntu server using Docker Compose, create a local repository on the same server, and push it to Gitea.

Step 1.1: Install and Run Gitea

Docker and Docker Compose were installed, Gitea and PostgreSQL containers were started, and Gitea responded on port 3000.

```
areeba@ubuntu:~/Gitea$ sudo docker compose ps && curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
NAME       IMAGE               COMMAND                  SERVICE   CREATED         STATUS         PORTS
gitea      gitea/gitea:latest  "/usr/bin/entrypoint…"   gitea     5 minutes ago   Up 5 minutes   22/tcp, 0.0.0.0:3000->3000/tcp, [::
]:3000->3000/tcp
gitea_db   postgres:17-alpine  "docker-entrypoint.s…"   db        5 minutes ago   Up 5 minutes   5432/tcp
Gitea HTTP status: 200
areeba@ubuntu:~/Gitea$
```

Figure 1: *task1_gitea_running.png*

Step 1.2: Create and Push the Initial Repository

A local Assignment01 repository with README.md (name and registration number) was created and pushed to Gitea using a personal access token.

```
areeba@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://areebahabib@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 266 bytes | 266.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To http://127.0.0.1:3000/areebahabib/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
areeba@ubuntu:~/Assignment01$
```

Figure 2: *task1_gitea_push.png*

Step 1.3: Verify the Gitea Repository

The repository was opened in Gitea to confirm it is public and the README is rendered.

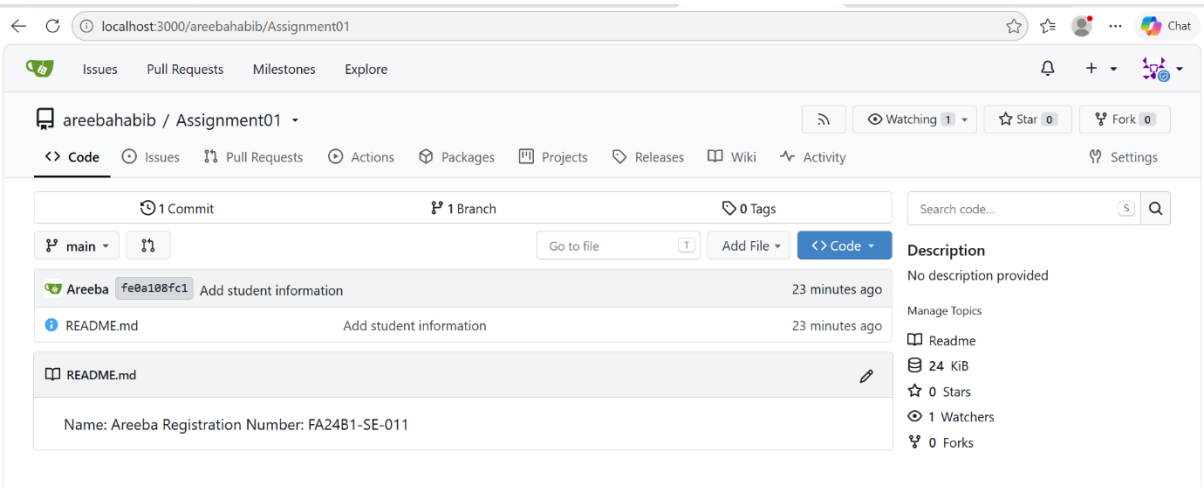


Figure 3: *task1_gitea_repository.png*

Task 2: Push the Same Repository to GitHub

Task meaning: Add GitHub as a second remote to the same local repository and push the main branch to it, so one repository is available on both Gitea and GitHub.

Step 2.1: Create the GitHub Repository and Verify Both Remotes

An empty public Assignment01 repository was created on GitHub and added as the github remote.

```
areeba@ubuntu:~/Assignment01$ git remote -v
gitea   http://areebahabib@127.0.0.1:3000/areebahabib/Assignment01.git (fetch)
gitea   http://areebahabib@127.0.0.1:3000/areebahabib/Assignment01.git (push)
github  https://github.com/areebabahabib/Assignment01.git (fetch)
github  https://github.com/areebabahabib/Assignment01.git (push)
```

Figure 4: *task2_remotes.png*

Step 2.2: Push to GitHub

The main branch was pushed to the github remote.

```
areeba@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': areebahabib
Password for 'https://areebahabib@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 266 bytes | 266.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/areebahabib/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
areeba@ubuntu:~/Assignment01$
```

Figure 5: *task2_github_push.png*

Step 2.3: Verify the GitHub Repository

The GitHub repository page was opened to confirm owner, name, public status and README.

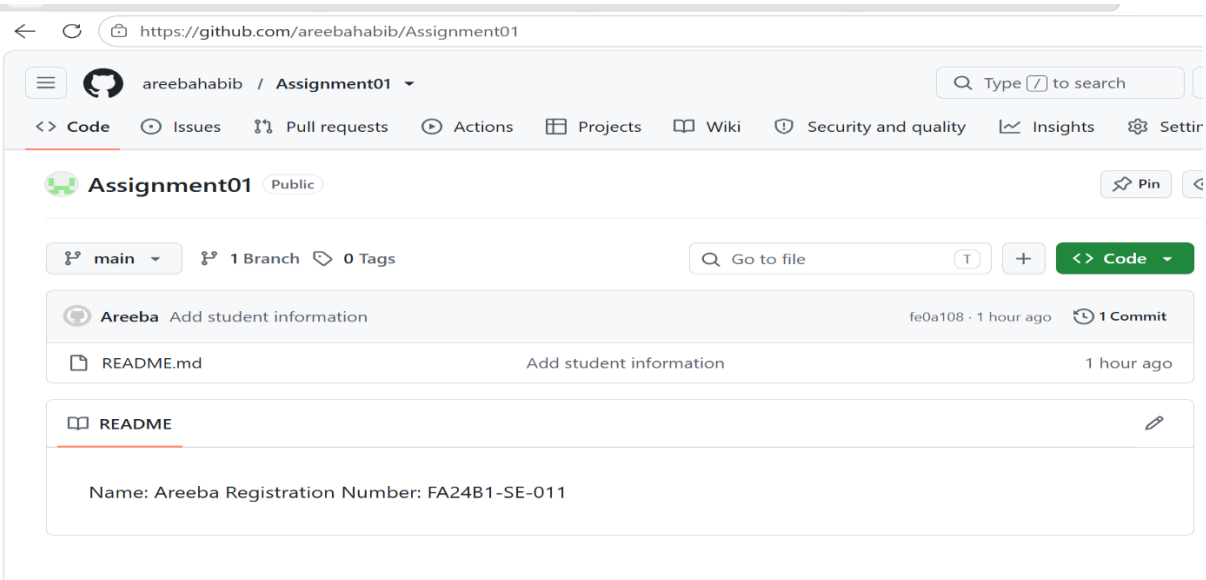


Figure 6: *task2_github_repository.png*

Task 3: Track Three Large Files with Git LFS

Task meaning: GitHub does not accept normal files above 100 MiB. Git LFS stores large files separately and keeps only small pointers in the repository. Three files larger than 100 MiB were tracked and pushed with LFS.

Step 3.1: Configure Git LFS

Git LFS was initialized and *.bin files were set to be tracked.

```
areeba@ubuntu:~/Assignment01$ cd ~/Assignment01
areeba@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
areeba@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.4.1 (GitHub; linux amd64; go 1.22.2)
areeba@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
areeba@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
areeba@ubuntu:~/Assignment01$
```

Figure 7: *task3_lfs_setup.png*

Step 3.2: Create and Verify Three LFS Files

Three .bin files (101M, 102M, 103M) were created, staged, and checked with Git LFS.

```
areeba@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
areeba@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
areeba@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
areeba@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
areeba@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
```

Figure 8: *task3_lfs_files.png*

Step 3.3: Commit and Push the LFS Files

The files were committed and pushed to GitHub; LFS uploaded the large objects.

```
areeba@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': areebahabib
Password for 'https://areebahabib@github.com':
Username for 'https://github.com': areebahabib
Password for 'https://areebahabib@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 1.7 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 796 bytes | 796.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/areebahabib/Assignment01.git
   fe0a108..e7e817b  main -> main
```

Figure 9: *task3_lfs_push.png*

Task 4: Create a Portfolio or CV with GitHub Pages

Task meaning: Build a portfolio website using HTML and CSS (About Me, Education, Skills, Projects) and publish it as a live site with GitHub Pages.

Step 4.1: Create and Push the Portfolio Repository

A public <username>.github.io repository was created with index.html and styles.css and pushed to main.

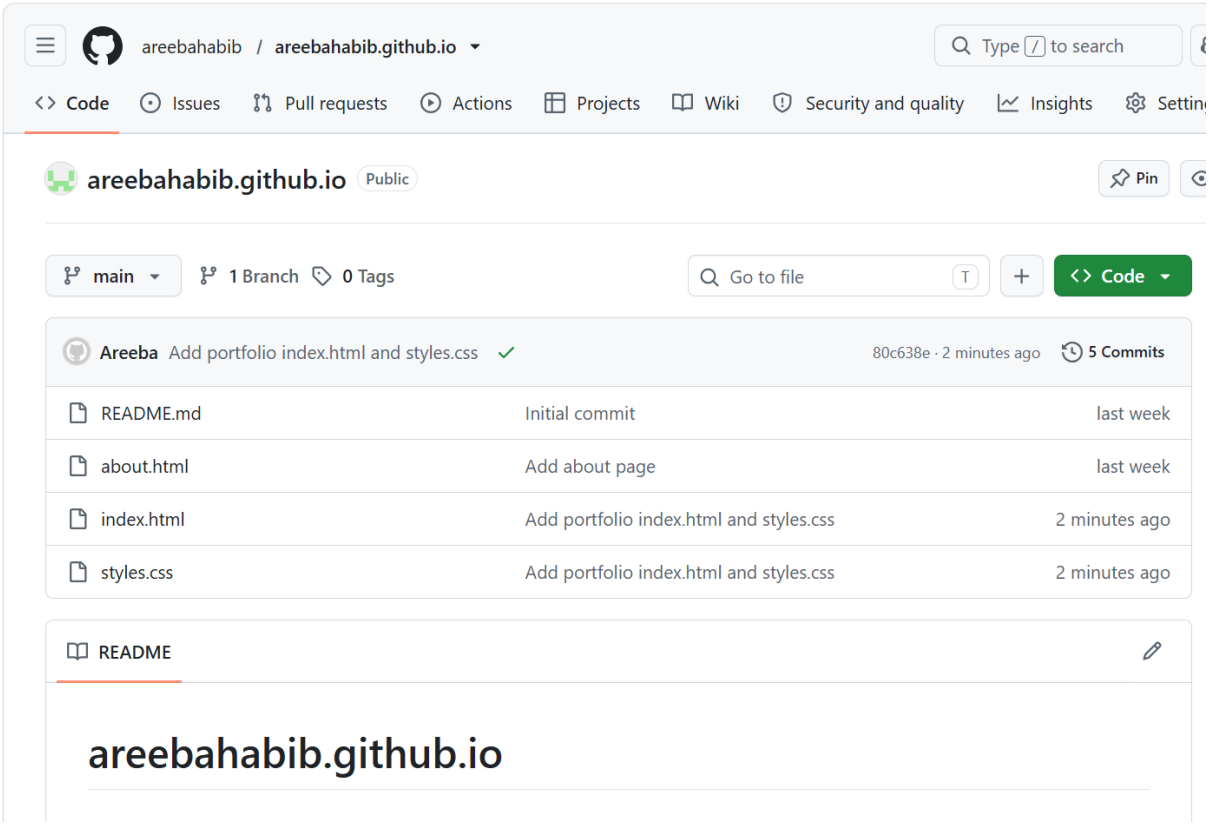


Figure 10: *task4_pages_repository.png*

Step 4.2: Publish with GitHub Pages

Pages was set to Deploy from a branch (main, / root) and the site went live.

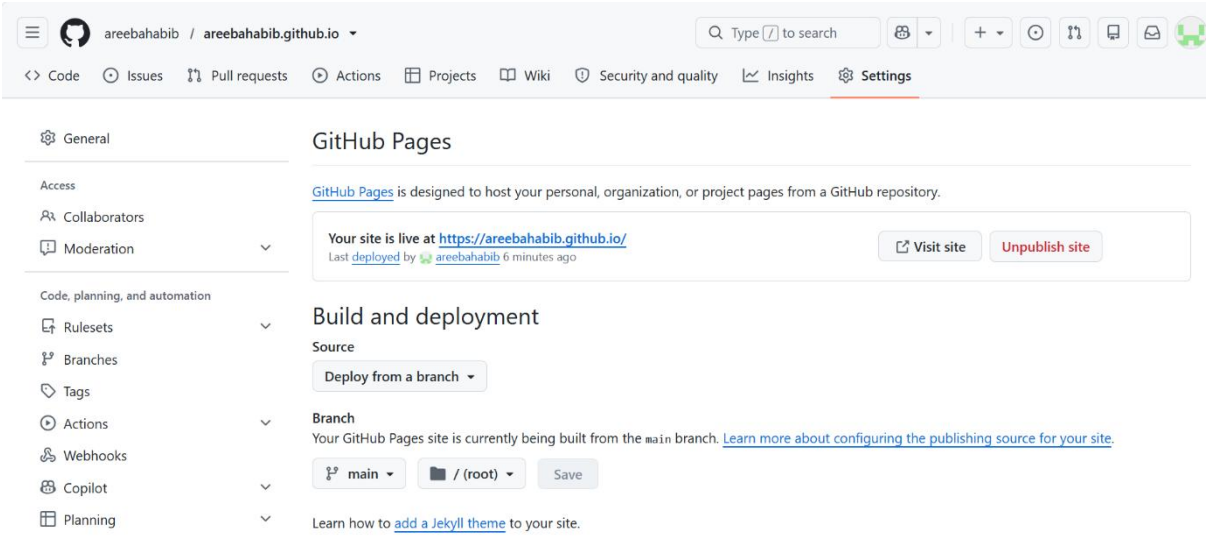


Figure 11: *task4_pages_deployment.png*

Step 4.3: Verify the Live Portfolio

The live site was opened in the browser.

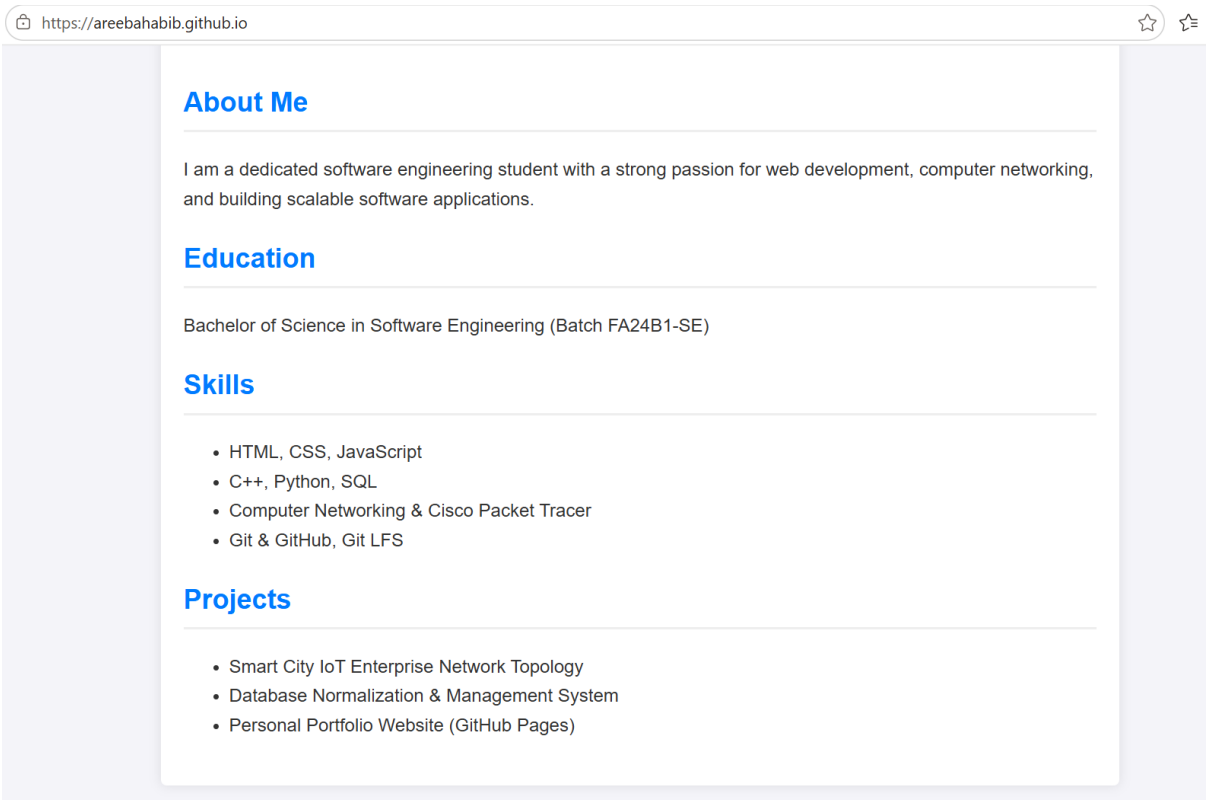


Figure 12: *task4_portfolio_live.png*

Submission Links

Item	Link
CC repository	https://github.com/areebahabib/CC
GitHub Assignment01	https://github.com/areebahabib/CC/tree/main/Assignments/Assignment01
Portfolio repository	https://github.com/areebahabib/areebahabib.github.io
Live portfolio	https://areebahabib.github.io/
Gitea Assignment01	http://localhost:3000/areebahabib/Assignment01